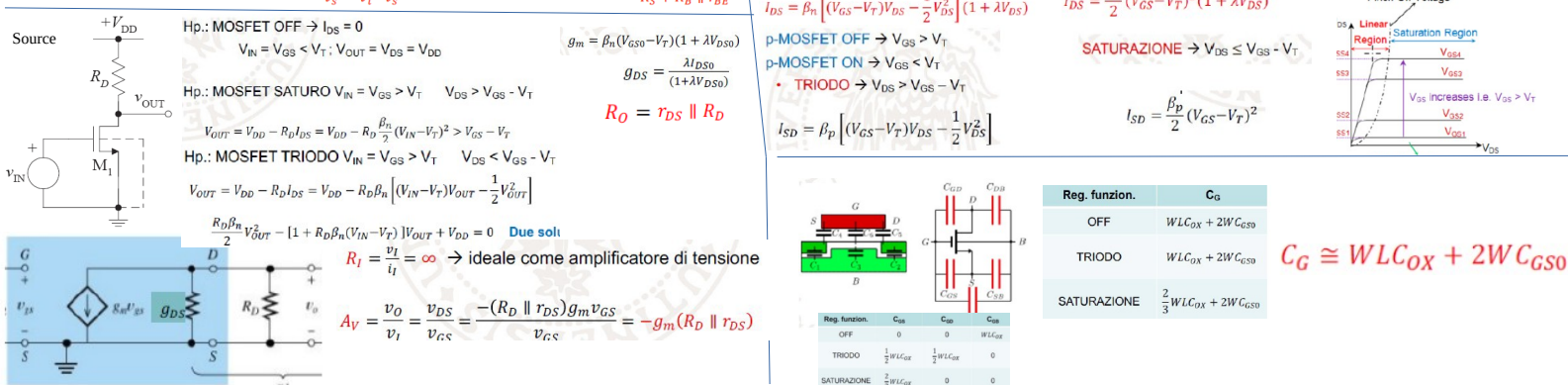
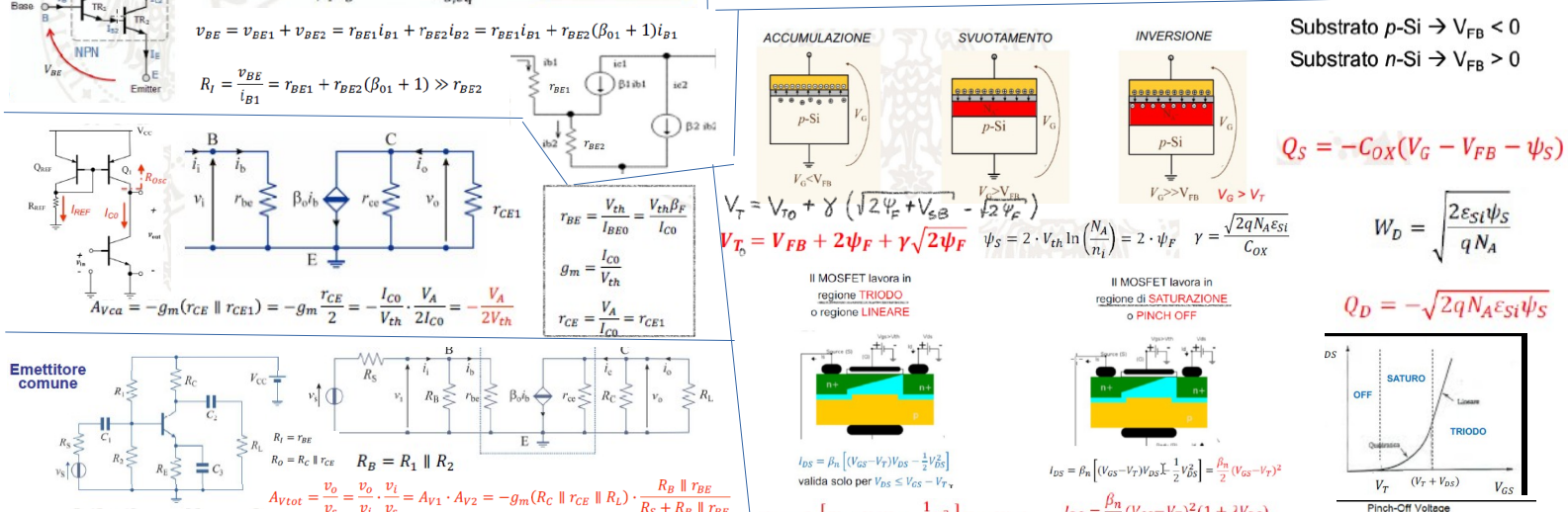
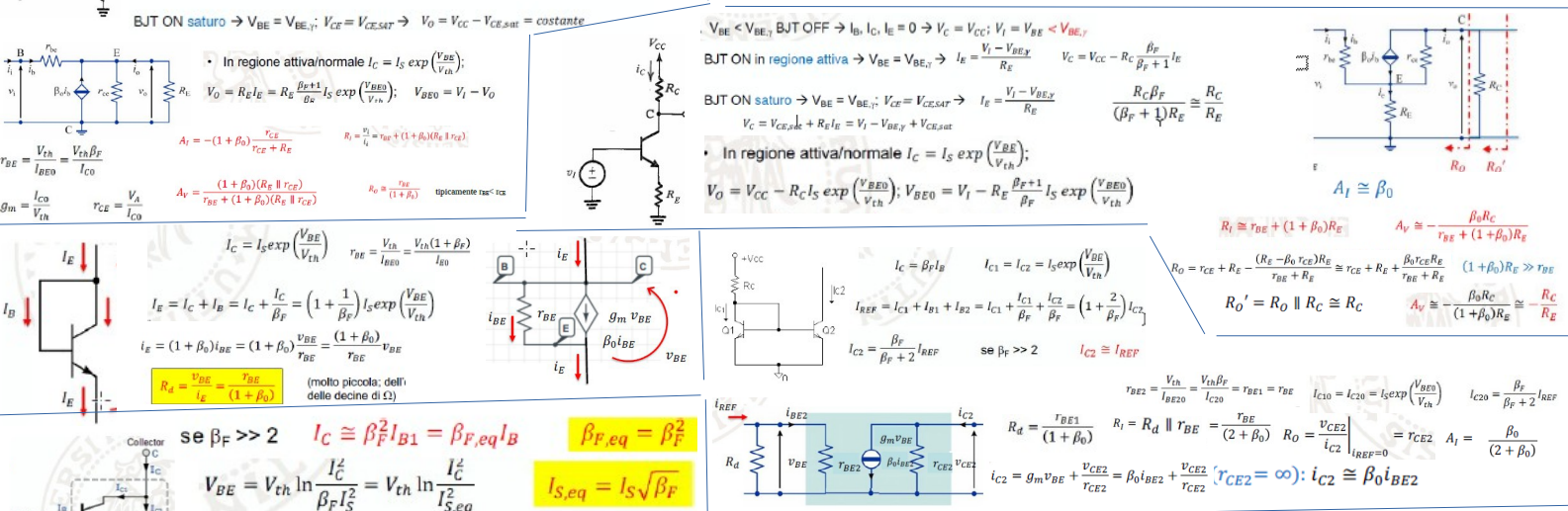
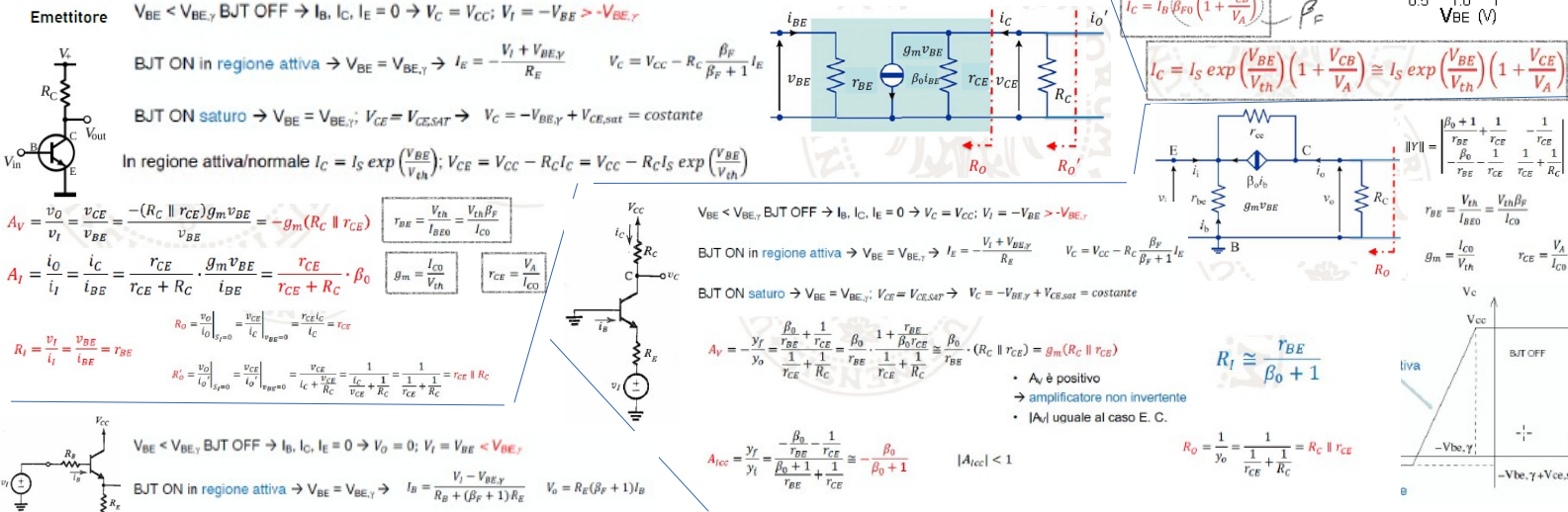
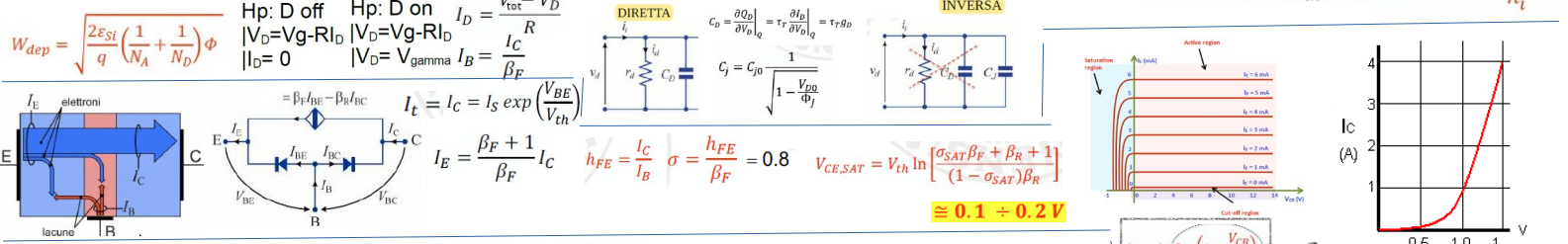


$J_{tot} = J_{tot,n} + J_{tot,p} = q(\mu_{n,n} + \mu_{p,p}) E + q D n \nabla n - q D p \nabla p \quad V_D = V_{th} \ln \left(\frac{I_D}{I_S} + 1 \right) \quad D_n / \mu_n = D_p / \mu_p = k_B T / q = V_{th} = 25.6 \text{ mV} \quad I_D = A \cdot J_S \cdot \left[\exp \left(\frac{V_D}{V_{th}} \right) - 1 \right] \quad \Phi_j = V_{th} \ln \frac{N_A N_D}{n_i^2}$



$R_1 = \frac{V_{th}}{I_{th}} = \infty$ $A_1 = \frac{V_{out}}{V_{in}} = \frac{1 + R_2(g_m(1 + \alpha) + g_{DS})}{1 + g_m(R_S \parallel r_{DS})} < 1$ $\alpha \approx 0 = \frac{g_m(R_S \parallel r_{DS})}{1 + g_m(R_S \parallel r_{DS})}$

$R_0 = \frac{1}{g_m(1 + \alpha) + g_{DS} + \frac{1}{R_S} + \frac{1}{r_{DS}} + \frac{1}{R_2}} \approx \frac{1}{g_m} \parallel \frac{1}{R_2} \parallel \frac{1}{r_{DS}}$

$I_D = \frac{\beta_n}{2}(V_{GS} - V_T)^2$

$V_{OUT} = R_S I_{DS0} = R_S \frac{\beta_n}{2}(V_{GS0} - V_T)^2$ $V_{GS0} = V_{IN} - V_{OUT}$

$V_{OUT} = V_{DD} - R_D I_{DS0} = V_{DD} - R_D \frac{\beta_n}{2}(V_{GS} - V_T)^2$ $V_{GS} = V_{IN} - V_{OUT}$

$g_m = \beta_n(V_{GS0} - V_T)(1 + \lambda V_{DS0})$

$g_{DS} = \frac{\lambda I_{DS0}}{(1 + \lambda V_{DS0})}$

$g_{mB} = -g_m \cdot \alpha$

$g_m = \beta_n(V_{GS0} - V_T)(1 + \lambda V_{DS0})$

$g_{DS} = \frac{\lambda I_{DS0}}{(1 + \lambda V_{DS0})}$

$g_{mB} = -g_m \cdot \alpha$

$R_0 = \frac{V_{out}}{I_o} = \frac{V_{out}}{I_{DS} + I_{DS0}} = \frac{V_{out}}{g_{DS}V_{GS} + \frac{V_{out}}{R_D} + \frac{V_{out}}{R_S}} = \frac{1}{\frac{g_{DS}}{1 + \lambda V_{DS0}} + \frac{1}{R_D} + \frac{1}{R_S}} = R_D \parallel R_S$ (come base comune)

Il MOSFET di destra è acceso per $V_{G1} = 3V$ $V_{G2} = 1V$

Il MOSFET di sinistra è acceso per $V_{G1} = V_{G2} > V_X + V_T$

$V_X < V_{GS} - V_T = 2.0V$ $V_{GS} = 1V$ $V_X = 3V$

$V_{GS} > V_X + V_T$ $V_X = 1V$ $V_{GS} = 3V$

$I_X = \frac{1}{2} \beta_n (V_{GS2} - V_T)^2 = \frac{1}{2} \beta_n (V_{GS} - V_X - V_T)^2$

$I_X = \beta_n ((V_{GS1} - V_T)V_{DS1} - \frac{1}{2} V_{DS1}^2) = \beta_n ((V_{GS1} - V_T)(V_X - V_{GS1}) - \frac{1}{2} (V_X - V_{GS1})^2)$

OFF $V_{GS} < V_{TN} \rightarrow V_i < V_{TN}$

TRIODO SATURO $V_{DS} < V_{GS} - V_{TN} \rightarrow V_{DS} < V_i - V_{TN}$ $V_{DS} > V_{GS} - V_{TN} \rightarrow V_{DS} > V_i - V_{TN}$

OFF $V_i < V_{DD} + V_{TP}$

TRIODO SATURO $V_{DS} > V_{GS} - V_{TP} \rightarrow V_{DS} > V_i - V_{TP}$ $V_{DS} < V_{GS} - V_{TP} \rightarrow V_{DS} < V_i - V_{TP}$

$I_{DSN} = \frac{\beta_n}{2}(V_{GSN} - V_{TN})^2 = I_{SDP} = \frac{\beta_p}{2}(V_{GSP} - V_{TP})^2$ $A_v = \frac{V_o}{V_i} = \frac{g_{m,n} \cdot g_{m,p}}{g_{m,n} + g_{m,p}} = \frac{2I_D(2 + \lambda V_{DS})}{2I_D(2 + \lambda V_{DS})}$

$I_1 = \frac{V_1}{Z_1} = \frac{V_1}{Z_1} = \frac{V_1}{Z_1}$

$I_2 = \frac{V_2}{Z_2} = \frac{V_2}{Z_2} = \frac{V_2}{Z_2}$

Matrice ammettenza

$G_{11} = \frac{I_1}{V_1} \Big|_{V_2=0}$ $G_{12} = \frac{I_1}{V_2} \Big|_{V_1=0}$ $V_1 \uparrow$

$G_{21} = \frac{I_2}{V_1} \Big|_{V_2=0}$ $G_{22} = \frac{I_2}{V_2} \Big|_{V_1=0}$

Matrice impedenza

$Z_{11} = \frac{V_1}{I_1} \Big|_{I_2=0}$ $Z_{12} = \frac{V_1}{I_2} \Big|_{I_1=0}$ $I_1 \uparrow$

$Z_{21} = \frac{V_2}{I_1} \Big|_{I_2=0}$ $Z_{22} = \frac{V_2}{I_2} \Big|_{I_1=0}$

Matrice ibrida

$h_{11} = \frac{V_1}{I_1} \Big|_{V_2=0}$ $h_{12} = \frac{V_1}{I_2} \Big|_{I_1=0}$ $I_1 \uparrow$

$h_{21} = \frac{I_2}{I_1} \Big|_{V_2=0}$ $h_{22} = \frac{I_2}{I_1} \Big|_{I_1=0}$

$I_1 = y_{11}V_1 + y_{12}V_2$

$I_2 = y_{21}V_1 + y_{22}V_2$

$\|Y'\| = \begin{bmatrix} y_{11}' & y_{12}' \\ y_{21}' & y_{22}' \end{bmatrix} = \begin{bmatrix} y_{11} + y_3 & y_{12} - y_3 \\ y_{21} - y_3 & y_{22} + y_3 \end{bmatrix}$

$V_1 = z_{11}I_1 + z_{12}I_2$ $V_2 = z_{21}I_1 + z_{22}I_2$

$\|Z'\| = \begin{bmatrix} z_{11}' & z_{12}' \\ z_{21}' & z_{22}' \end{bmatrix} = \begin{bmatrix} z_{11} - z_3 & z_{12} + z_3 \\ z_{21} + z_3 & z_{22} - z_3 \end{bmatrix}$

$V_2^B = m_{11}^B V_1^B - m_{12}^B I_1^B$

$I_2^B = m_{21}^B V_1^B - m_{22}^B I_1^B$

$V_2^A = m_{11}^A V_1^A - m_{12}^A I_1^A$

$I_2^A = m_{21}^A V_1^A - m_{22}^A I_1^A$

Amplificatore unilatero ($h_{re}=0$)

$A_{vcc} = h_{fe} A_{vce} = -\frac{h_{fe}}{D_n} = -\frac{h_{fe}}{h_{ie} h_{oe} + h_{re} h_{fe}} = \frac{1}{h_{ie} h_{oe} + h_{re} h_{fe}}$

BJT in frequenza

$\omega_T = \frac{\beta_0}{r_{BE}(C_{BE} + C_{BC})} = \frac{g_m}{C_{BE} + C_{BC}}$

Mosfet in frequenza

$\|Y'\| = \begin{bmatrix} y_{11}' & y_{12}' \\ y_{21}' & y_{22}' \end{bmatrix} = \begin{bmatrix} g_{m1} + sC_{GS} & -sC_{GD} \\ g_m - sC_{GD} & g_{DS} + sC_{GD} \end{bmatrix}$

$\omega_T = \frac{g_m}{C_{GS} + C_{GD}}$ $|A_{vcc}(\omega)| \approx \frac{g_m}{\omega(C_{GS} + C_{GD})}$

MOSFET: C_{GS} e C_{GD} Costanti $\rightarrow \omega_T \propto \sqrt{I_{DS0}}$ (dipende dal punto di lavoro)

BJT: tipicamente $C_{BE} \gg C_{BC} \rightarrow \omega_T = \frac{g_m}{C_{BE} + C_{BC}} = \frac{g_m}{C_{BE}} = \frac{1}{\tau_{BE}}$ (non dipende dal punto di lavoro)

n-MOSFET:

OFF $V_{GS} \leq V_T$ $I_{DS} = 0$

TRIODO $V_{GS} > V_T$ $V_{DS} \leq V_{GS} - V_T$ $I_{DS} = \beta_n \left[(V_{GS} - V_T) V_{DS} - \frac{V_{DS}^2}{2} \right] (1 + \lambda \cdot V_{DS})$

SATURO $V_{GS} > V_T$ $V_{DS} > V_{GS} - V_T$ $I_{DS} = \frac{\beta_n}{2} (V_{GS} - V_T)^2 (1 + \lambda \cdot V_{DS})$

p-MOSFET:

OFF $V_{SG} \leq |V_T|$ $I_{SD} = 0$

TRIODO $V_{SG} > |V_T|$ $V_{SD} \leq V_{SG} - |V_T|$ $I_{SD} = \beta_p \left[(V_{SG} - |V_T|) V_{SD} - \frac{V_{SD}^2}{2} \right] (1 + \lambda \cdot V_{SD})$

SATURO $V_{SG} > |V_T|$ $V_{SD} > V_{SG} - |V_T|$ $I_{SD} = \frac{\beta_p}{2} (V_{SG} - |V_T|)^2 (1 + \lambda \cdot V_{SD})$

Emettitore comune

$A_v = \frac{\beta_o (R_C \parallel r_{ce})}{r_{be} + (\beta_o + 1)(R_E \parallel r_{ce})}$

$A_i = \frac{\beta_o r_{ce}}{r_{ce} + R_C}$

$R_{in} = r_{be}$

$R_{out} = R_C \parallel r_{CE}$

Collettore comune

$\frac{(\beta_o + 1)(R_E \parallel r_{ce})}{r_{be} + (\beta_o + 1)(R_E \parallel r_{ce})}$

$-(\beta_o + 1) \frac{r_{ce}}{r_{ce} + R_E}$

$r_{be} + (\beta_o + 1)(R_E \parallel r_{ce})$

$\frac{r_{BE}}{\beta_o + 1}$

Base comune

$\frac{\beta_o (R_C \parallel r_{ce})}{r_{be}}$

$-\frac{\beta_o r_{ce}}{R_C + (\beta_o + 1)r_{ce}}$

$\frac{r_{BE}}{\beta_o + 1}$

$R_C \parallel r_{CE}$

p-MOSFET

Il p-MOSFET al piccolo segnale si comporta in modo analogo al suo compare con qualche differenza sui segni.

Regione triodo $g_m = \beta_p |V_{GS0} - V_T|$ $g_{DS} = \beta_p |V_{GS0} - V_T - V_{DS0}|$

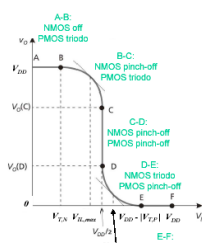
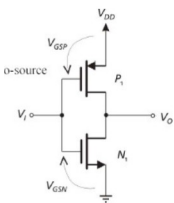
Regione saturazione $g_m = \beta_p |V_{GS0} - V_T|$ $g_{DS} = 0$

$I_t = I_s \left[\exp\left(\frac{V_{BE}}{V_{th}}\right) - \exp\left(\frac{V_{CB}}{V_{th}}\right) \right]$

DIRETTA: $C_{BE} = \frac{dQ_B}{dV_{BE}} \rightarrow Q_B = \tau_F \cdot I_t$ $\tau_F = \frac{W_{BASE}^2}{2 \cdot D_n}$

INVERSA: $C_{BC} = \frac{dQ_n}{dV_{CB}} = \frac{C_{J0}}{\sqrt{1 - \frac{V_{CB}}{V_s}}}$

Inverter CMOS



$$V_{LT} = \frac{V_{TN} + \sqrt{\frac{\beta_p}{\beta_n}}(V_{DD} + V_{TP})}{1 + \sqrt{\frac{\beta_p}{\beta_n}}}$$

$$t_{SAT} = \frac{2C_L V_{TN}}{\beta_n (V_{DD} - V_{TN})^2} \quad t_f = t_{SAT} + t_{TR} = \frac{2C_L}{\beta_n (V_{DD} - V_{TN})} \left[\frac{V_{TN}}{V_{DD} - V_{TN}} + \frac{1}{2} \ln \frac{2(V_{DD} - V_{TN}) - V_{OLmax}}{V_{OLmax}} \right]$$

$$t_{pr} = t_{pr0} + \eta \cdot t_{ri}$$

$$\eta = 0.25 \text{ V}$$

$$t_r = t_{SAT} + t_{TR} = \frac{2C_L}{\beta_p (V_{DD} + V_{TP})} \left[\frac{-V_{TP}}{V_{DD} + V_{TP}} + \frac{1}{2} \ln \frac{2(V_{DD} + V_{TP}) - (V_{DD} - V_{OHmin})}{V_{DD} - V_{OHmin}} \right]$$

$$F(r) = \frac{1}{1-r} \left[r + \frac{1}{2} \ln(19-20r) \right]$$

$$t_f = \frac{2C_L}{\beta_n V_{DD}} F\left(\frac{V_{TN}}{V_{DD}}\right) \quad t_r = \frac{2C_L}{\beta_p V_{DD}} F\left(\frac{|V_{TP}|}{V_{DD}}\right)$$

$$t_p = \frac{t_f + t_r}{2} = \frac{C_L}{V_{DD}} \left(\frac{1}{\beta_n} + \frac{1}{\beta_p} \right) F\left(\frac{V_T}{V_{DD}}\right)$$

$$P_{din} = C_L V_{DD}^2 f_{0 \rightarrow 1}$$

$$I_{peak} = \frac{\beta_n}{2} (V_{LT} - V_T)^2 = \frac{\beta_n}{2} \left(\frac{V_{DD}}{2} - V_T \right)^2$$

$$P_{SC} = \frac{4t_c}{3} I_{peak} \left(\frac{V_{DD}}{2} - V_T \right) \cdot f$$

$$C_{inv} = C_{Gn} + C_{Gp} = (W_n + W_p) L_{MIN} C_{OX} + 2(W_n + W_p) C_{GS0}$$

$$C_{inv} = S_n (1 + \alpha) C_{M1}$$

$$C_{M1} = L_{MIN}^2 C_{OX} + 2L_{MIN} C_{GS0}$$

$$K_{eq} = 0.6$$

$$C_{DBeq} = K_{eq} W L_S C_{j0}$$

$$C_{GDeq} = 2(W_n + W_p) C_{GS0} + \frac{1}{2} (W_n + W_p) L_{MIN} C_{OX} R_{DD}$$

$$C_L = C_{GDeq} + C_{DBeq} + C_W + C_{inv2} =$$

$$= 2(W_n + W_p) C_{GS0} + \frac{1}{2} (W_n + W_p) L_{MIN} C_{OX} R_{DD} + K_{eq} (W_n + W_p) L_S C_{j0} + C_W + 2(W_{n2} + W_{p2}) C_{GS0} + (W_{n2} + W_{p2}) L_{MIN} C_{OX}$$

$$C_L = (1 + \alpha) C_n + C_W$$

$$t_p = \frac{t_r + t_f}{2} = \frac{C_L}{V_{DD}} \left(\frac{1}{\beta_n} + \frac{1}{\beta_p} \right) F\left(\frac{V_T}{V_{DD}}\right) = \frac{(1 + \alpha) C_n + C_W}{V_{DD}} \left(\frac{1}{\beta_n} + \frac{1}{\beta_p} \right) F\left(\frac{V_T}{V_{DD}}\right)$$

ostituzioni
(si)

$$\alpha_{opt} = \sqrt{\epsilon}$$

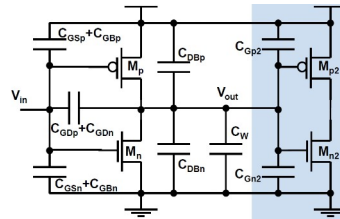
$$t_{popt} = \frac{(1 + \sqrt{\epsilon})^2 C_n}{\beta_n V_{DD}} F\left(\frac{V_T}{V_{DD}}\right)$$

$$t_{pi} = \frac{C_{i+1}}{C_i} \cdot t_{p0} = U \cdot t_{p0}$$

$$t_{pN} = \frac{C_L}{C_N} \cdot t_{p0} = \frac{C_L}{U(N-1)C_1} \cdot t_{p0} \quad (\text{tempo di ritardo dell'ultimo buffer})$$

$$N_{opt} = \frac{\ln\left(\frac{C_L}{C_{in}}\right)}{\ln U} = \frac{\ln X}{\ln U}$$

$$t_{popt} = N_{opt} \cdot U_{opt} \cdot t_{p0} = \ln\left(\frac{C_L}{C_{in}}\right) \cdot e \cdot t_{p0} \quad N_{opt} = \ln X = \ln\left(\frac{C_L}{C_{in}}\right)$$



Pilotaggio di grandi capacità

$$t_p = \frac{C_L}{C_{in}} \cdot t_{p0}$$

$$t_{p0} = \frac{2(1 + \alpha) C_{M1}}{\beta_n V_{DD}} F\left(\frac{V_T}{V_{DD}}\right)$$

$$C_{in2} = S_n (1 + \alpha) C_{M1} = S(1 + \epsilon) C_{M1} = S C_{in}$$

$$X = \frac{C_L}{C_{in}}$$

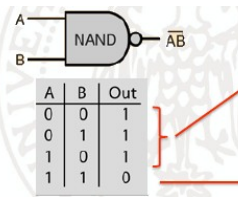
$$t_p = t_{p1} + t_{p2} = \frac{C_{in2}}{C_{in}} \cdot t_{p0} + \frac{C_L}{C_{in2}} \cdot t_{p0} = t_{p0} \cdot \left(\frac{C_{in2}}{C_{in}} + \frac{C_L}{C_{in2}} \right) = t_{p0} \cdot \left(S + \frac{C_L}{C_{in}} \cdot \frac{C_{in}}{C_{in2}} \right)$$

$$S_{opt} = \sqrt{X}$$

$$t_{popt} = 2t_{p0} \cdot \sqrt{X} = 2t_{p0} \cdot \sqrt{\frac{C_L}{C_{in}}}$$

$$\alpha = \frac{S_p}{S_n} \quad \alpha = \frac{\beta_n'}{\beta_p'} \quad \text{per } t_r = t_f$$

Porte logiche statiche

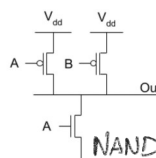


PULL-UP (PU)
responsabile di
generare $F = 1$

$$PU(\bar{A}, \bar{B}) = F(A, B)$$

PULL-DOWN (PD)
responsabile di
generare $F = 0$

$$PD(A, B) = \bar{F}(A, B)$$



$$I = (S_1 + S_2) \beta_n' \left[(V_{GS} - V_T) V_{DS} - \frac{1}{2} V_{DS}^2 \right]$$

$$\text{UNICO MOSFET con } S_{eq} = (S_1 + S_2)$$

→ il risultato è identico se i MOSFET sono in saturazione!!

In serie di MOSFET

$$I = S_{eq} [g(V_{GB}, V_{DB}) - g(V_{GB}, V_{SB})] = S_{eq} \beta_n' \left[(V_{GS} - V_{TN}) V_{DS} - \frac{1}{2} V_{DS}^2 \right] I = \frac{S_{eq} \beta_n'}{2} (V_G - V_{TN})^2$$

$$S_{eq} = \frac{S_1 S_2}{S_1 + S_2}$$

$$C_L = k C_{in,v} + C_W + S_n (1 + m \alpha_{NAND}) C_{p1}$$

$$C_{p1} = L_{MIN} \left[2C_{GS0} + \frac{1}{2} L_{MIN} C_{OX} R_{DD} + K_{eq} L_S C_{j0} \right]$$

$$P_{din} = C_L V_{DD}^2 f_{in}$$

$$f_{0 \rightarrow 1} = f_{CK} P_{0 \rightarrow 1}$$

$$P_{0 \rightarrow 1} = P_0 \cdot P_1 = (1 - P_1) \cdot P_1 = P_0 \cdot (1 - P_0)$$

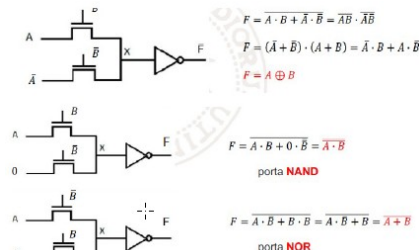
Pass-transistor in serie

$$R_{eq} = \frac{2V_{DD}}{\beta_n (V_{DD} - V_{TN0})^2 + \beta_p (V_{DD} - |V_{TP0}|)^2}$$

$$t_r = R_{eq} C \cdot \frac{n(n-1)}{2}$$

$$\text{è circa proporzionale a } n^2 \quad m_{opt} = \sqrt{\frac{2T_{buffer}}{2.3 \cdot R_{eq} C}}$$

$$t_{r,x} \approx 2.3 \cdot t_n$$



SCARICA di C_L → attraverso 2 MOSFET in parallelo

→ il caso peggiore è quando la scarica è attraverso

1 solo MOSFET

→ identico al caso dell'inverter

$$t_{f,NOR} = \frac{2C_L}{\beta_p' S_{n,NOR} V_{DD}} F\left(\frac{V_T}{V_{DD}}\right)$$

$$S_{n,NOR} \rightarrow \text{dimensionamento n-MOSFET}$$

$$t_{r,NOR} = \frac{4C_L}{\beta_p' \alpha_{NOR} S_{n,NOR} V_{DD}} F\left(\frac{V_T}{V_{DD}}\right)$$

$$t_{f,NAND} = \frac{2F\left(\frac{V_T}{V_{DD}}\right)}{\beta_p' V_{DD}} \cdot \left[\frac{m}{S_n} \cdot (k C_{in,v} + C_W) + m \cdot (1 + m \alpha_{NAND}) C_{p1} \right]$$

EFFETTO DEL CARICO
dipende dal FAN OUT !!

EFFETTO DEL SELF-LOADING
dipende dal FAN IN ! non dipende da S_n !

Logiche dinamiche CMOS

$$P_{din} = C_L V_{DD}^2 f_{CK} P_0$$

$$P_{0 \rightarrow 1} = \frac{N_0}{2N}$$

